UNIVERSITY OF MANITOBA
Department of Mathematics

COURSE: *Math 1210* EXAMINER: *Romain Branchereau*
DATE: *June 19, 2023* TIME: *2 hours*
EXAM: *Final exam*

First name (please write as legibly as possible within the boxes)

Last name

Student ID number

Grading table

Question:	1	2	3	4	5	6	7	8	9	10	Total
Points:	8	6	10	14	9	7	6	12	7	15	94
Score:											

Instructions

- Answer all questions on the exam paper in the space provided beneath the question. Unless otherwise specified, your answer must be justified. Unjustified answers will receive little or no credit.
- If you need more space, continue on the back of the same page, clearly indicating that your work is to be continued.
- The value of each question is indicated beside the statement of the question. The total value of all questions is 94 points.
- No texts, notes, or other aids are permitted. There are no calculators, cellphones, or any other electronic device permitted.
- Do not remove staples or any pages from the exam except the last blank page.
- If you have any questions, feel free to ask me.
- Good Luck!

1. (8 points) Prove by induction that

$$\begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}^n = \begin{pmatrix} 1-2n & n \\ -4n & 1+2n \end{pmatrix}$$

for any $n \geq 1$.

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2. (6 points) Find the Cartesian form of

$$z = \frac{i^{62}(\sqrt{2}+\sqrt{6}i)^5}{2^6(-\sqrt{6}-\sqrt{2}i)}.$$

Simplify as much as possible.

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3. Consider the polynomial $P(x) = x^3 + 4x^2 + 7x + 6$.

   (a) (4 points) Use the rational root theorem to list all possible rational roots. Simplify the list as much possible using the other theorems that we have seen in class.

   (b) (6 points) Find all the zeros of $P(x)$.

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4. (*Part (d) is on the next page)* - Consider the vectors $\mathbf{u} = \langle 1, m, 2\rangle$,$\mathbf{v} = \langle m, 2m, 2\rangle$ and $\mathbf{w} = \langle 0, -3, 1\rangle$ where m is a real number.

   (a) (2 points) For which value(s) of m are $\mathbf{u}$ and $\mathbf{u} \times \mathbf{v}$ perpendicular?

   (b) (2 points) For which value(s) of m are $\mathbf{v}$ and $\mathbf{w}$ perpendicular?

   (c) (4 points) For which value(s) of m are the three vectors linearly dependent?

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(d) (6 points) Now take $m = 1$. Write the vector $\langle 1, -12, 5 \rangle$ as a linear combination of $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$.

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5. Consider the plane Π: $2x - 3y + z = 5$, the line L: $x = 1 + t,\ y = 2 + t,\ z = 9 + t$ and the point $P(1, -1, 2)$.

   (a) (3 points) Show that the line L is contained in the plane Π.

   (b) (6 points) Find an equation of the plane passing through the point P and intersecting the plane Π along the line L.

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6. (7 points) Consider the system

$$\begin{aligned} x - y + 2z &= 1 \\ x + 2y + z &= -1 \\ 2x - y + 3z &= 0. \end{aligned}$$

It is given to you that the system has a unique solution (x, y, z). Use Cramer's rule to find z only. (You do **not** need to find x and y).

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7. (6 points) Consider the invertible matrix

$$A = \begin{pmatrix} 2 & -1 & -3 \\ 2 & 0 & 1 \\ 1 & -2 & -1 \end{pmatrix}.$$

Use the adjoint method to find the entry in the second column and third row of A^{-1}.

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8. Consider the system

$$\begin{aligned} -x_1 + x_2 - 3x_3 + x_4 &= a \\ -x_1 + 2x_2 - 4x_3 + 2x_4 &= a^2 \\ x_1 + 2x_3 &= a^3 \end{aligned}$$

where a is a real number.

(a) (3 points) For which value(s) of a does the system have a unique solution?

(b) (9 points) Use Gauss-Jordan elimination to find the basic solutions of the system when $a = 0$.

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9. (7 points) Consider the linear transformation $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ sending $\langle 3, 2\rangle$ to $\langle 8, 0\rangle$, and $\langle 1, 0\rangle$ to $\langle 2, -2\rangle$. First, use that T is linear to find the image of $\langle 0, 1\rangle$. Then, find the associated matrix.

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10. (*Part (c) is on the next page)* - Consider the linear transformation T with associated matrix

$$A = \begin{pmatrix} 3 & 1 & 5 \\ 1 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix}.$$

(a) (2 points) Find the image of $\langle 2, -3, 1 \rangle$ by T.

(b) (7 points) Find the eigenvectors to the eigenvalue $\lambda = 2$.

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(c) (6 points) Find the other eigenvalues. You do **not** need to find the corresponding eigenvectors.

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